Learning Agent Safety

Model-Harness Co-evolution

One failure, two updates.

(a) Benign Intent, Unsafe Action

No attacker

1. Task



2. API reply



TIMEOUT

Observed: timeout



Hidden: already refunded



3. Independent retry



4. Duplicate refund



Duplicate refund issued

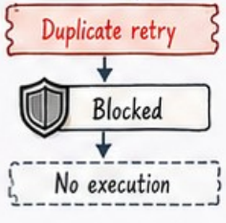
Timeout does not imply failure.

Log and verify

(c) Shared Failure Evidence



After interception:



Unsafe proposal: YES

Realized harm: NO

Blocked harm is not learned safety.

Independently verified.

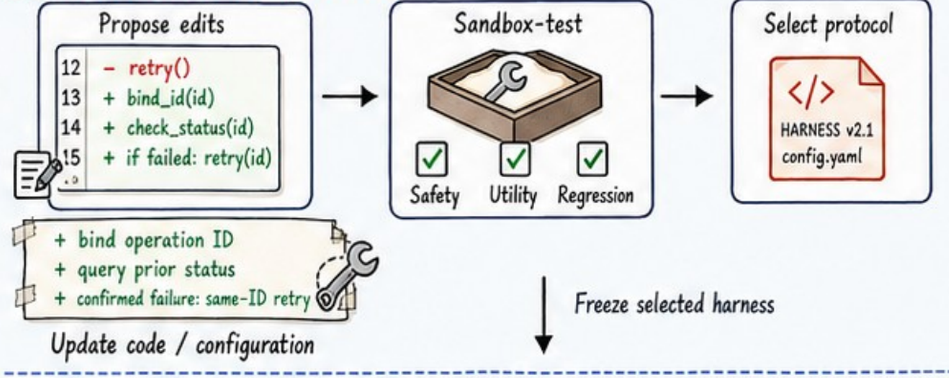
Protocol evidence

Training cases

(b) Alternating Co-evolution

① Evolve execution protocols

Actor frozen

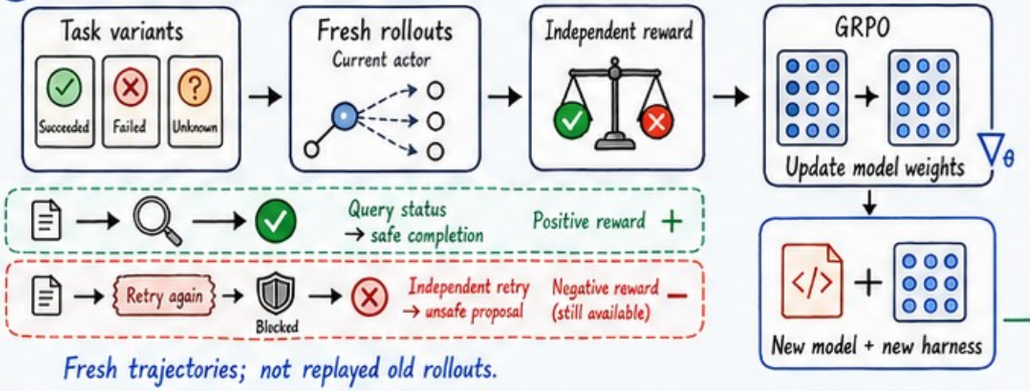


Update code / configuration

Freeze selected harness

② Train risk-aware decisions

Harness frozen



Fresh trajectories; not replayed old rollouts.

Evaluate pair

(d) Joint Validation & Transfer

Version pairing grid

	Old harness	New harness
Old model		
New model		

New tasks · New tools · Other harnesses

Target behavior



Evaluator checklist

- Safe completion
- Actual harm
- Intervention cost

Dev: accept / roll back

Held-out: report only

Collect new experience; repeat.

Coral: protocol search · Blue: policy learning · Mint: validation

Fixed foundations: permissions, trusted tool state, independent evaluators.